

Applied NLP

DATA 641

Lecture 2 — NLP Pipeline; Advanced Pre-processing

*Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems 1st Edition, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, O' Reilly.

Advanced Pre-processing

Main idea

Imagine we are asked to develop a system to identify person and organization names in our company's collection of one million documents. The common pre-processing steps we discussed earlier may not be relevant in this context. Identifying names requires us to be able to do Part-of-speech (POS) tagging, as identifying proper nouns can be useful in identifying person and organization names.

POS Tagging: What is it?

- Given a string:

```
This is a simple sentence
```

- Identify parts of speech (syntactic categories):

```
This/DET is/VB a/DET simple/ADJ sentence/NOUN
```

POS Tagging: Why Care?

POS tagging is a first step towards syntactic analysis (which in turn, is often useful for semantic analysis)

- Simpler models and often faster than full parsing, but sometimes enough to be useful.
- For example, POS tags can be useful features in text classification or word sense disambiguation

POS Tagging: Why is it Difficult?

- Ambiguity:
 - glass of water/NOUN vs. water/VERB the plants
 - lie/VERB down vs. tell a lie/NOUN
 - wind/VERB down vs. a mighty wind/NOUN
 - How about "time flies like an arrow"?
- Context sensitivity
- Language variations
- Overlapping categories
- Lack of agreement
- Out-of-Vocabulary Words

The Penn Treebank Dataset

The Penn Treebank, in its eight years of operation (1989-1996), produced:

- ~7 million words of part-of-speech tagged text
- 3 million words of skeletally parsed text
- over 2 million words of text parsed for predicate-argument structure
- and 1.6 million words of transcribed spoken text annotated for speech disfluencies

The material annotated includes: IBM computer manuals, nursing notes, Wall Street Journal articles, and transcribed telephone conversations, among others

The Penn Treebank Dataset

Tag	Description	Example	Tag	Description	Example
CC	coordin. conjunction	<i>and, but, or</i>	SYM	symbol	<i>+, %, &</i>
CD	cardinal number	<i>one, two</i>	TO	“to”	<i>to</i>
DT	determiner	<i>a, the</i>	UH	interjection	<i>ah, oops</i>
EX	existential ‘there’	<i>there</i>	VB	verb base form	<i>eat</i>
FW	foreign word	<i>mea culpa</i>	VBD	verb past tense	<i>ate</i>
IN	preposition/sub-conj	<i>of, in, by</i>	VBG	verb gerund	<i>eating</i>
JJ	adjective	<i>yellow</i>	VCN	verb past participle	<i>eaten</i>
JJR	adj., comparative	<i>bigger</i>	VBP	verb non-3sg pres	<i>eat</i>
JJS	adj., superlative	<i>wildest</i>	VBZ	verb 3sg pres	<i>eats</i>
LS	list item marker	<i>1, 2, One</i>	WDT	wh-determiner	<i>which, that</i>
MD	modal	<i>can, should</i>	WP	wh-pronoun	<i>what, who</i>
NN	noun, sing. or mass	<i>llama</i>	WP\$	possessive wh-	<i>whose</i>
NNS	noun, plural	<i>llamas</i>	WRB	wh-adverb	<i>how, where</i>
NNP	proper noun, sing.	<i>IBM</i>	\$	dollar sign	<i>\$</i>
NNPS	proper noun, plural	<i>Carolinas</i>	#	pound sign	<i>#</i>
PDT	predeterminer	<i>all, both</i>	“	left quote	<i>‘ or “</i>
POS	possessive ending	<i>’s</i>	”	right quote	<i>’ or ”</i>
PRP	personal pronoun	<i>I, you, he</i>	(	left parenthesis	<i>[, (, {, <</i>
PRP\$	possessive pronoun	<i>your, one’s</i>	)	right parenthesis	<i>],), }, ></i>
RB	adverb	<i>quickly, never</i>	,	comma	<i>,</i>
RBR	adverb, comparative	<i>faster</i>	.	sentence-final punc	<i>. ! ?</i>
RBS	adverb, superlative	<i>fastest</i>	:	mid-sentence punc	<i>: ; ... - -</i>
RP	particle	<i>up, off</i>			

Methods to Train POS Models

- Rules-based:
 - start with a dictionary of words and possible tags
 - assign all possible tags to words using the dictionary
 - write rules by hand to selectively remove tags
 - stop when each word has exactly one (presumably correct) tag

Need a rules-dictionary! -e.g., EnCG ENGTWOL tagger

Methods to Train POS Models

- Probabilistic models

For tokens $w_1, \dots, w_n$, find the most probable corresponding sequence of possible tags $t_1, \dots, t_n$

Find the best tag sequence T for an *untagged* sentence W :

$$\arg \max_T P(T | W)$$

- Models based on machine learning and deep learning

Methods based on ML and DL have not shown to be superior to Stochastic/Probabilistic methods in PoS tagging. However, there is always the computational complexity that one needs to consider.

POS Tagging: What are the available tools?

- Generally do not have to develop our own POS-tagging solutions.
- Pre-trained and readily usable POS taggers are implemented in NLP libraries such as NLTK , spaCy , etc.

```
In [1]: import nltk
        from nltk.tokenize import word_tokenize

        corpus = "Charles Spencer Chaplin was born in London England on April 16th 1889"

        print("POS Tagging using NLTK:")
        print(nltk.pos_tag(word_tokenize(corpus)))
```

POS Tagging using NLTK:

```
[('Charles', 'NNP'), ('Spencer', 'NNP'), ('Chaplin', 'NNP'), ('was', 'VBD'), ('born', 'VBN'), ('in', 'IN'), ('London', 'NNP'), ('England', 'NNP'), ('on', 'IN'), ('April', 'NNP'), ('16th', 'CD'), ('1889', 'CD')]
```

Examples of Other Tagging Tasks

- Named entity recognition:
 - e.g., label words as belonging to persons, organizations, locations, or none of the above:
 - Barack/PER Obama/PER spoke/NON from/NON the/NON White/LOC House/LOC today/NON
- Information field segmentation:
 - Given specific type of text (classified advert, bibliography entry), identify which words belong to which "fields"(price/size/location, author/title/year)